

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0011070622617831403
rotor delay [s]: 0.180077200586
gimbal delay [s]: 0.041131384435
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 5.4067049689226105e-17
common-scale FD null norm: 1.18854240480888e-11

Dimensionless coordinate

log_mass_scale	-0.227603358078
log_second_moment_mode_xx	-1.7106270051
log_second_moment_mode_yy	0.640499651711
log_second_moment_mode_zz	2.85091253292
log_second_moment_mode_xy	-0.604034159084
log_second_moment_mode_xz	1.95395232298
log_second_moment_mode_yz	0.345816229579
dimensionless_cog_displacement_x	-0.0379932207153
dimensionless_cog_displacement_y	0.0324504971712
dimensionless_cog_displacement_z	-0.238565557797
log_force_effectiveness_1	-0.386538398272
log_force_effectiveness_2	-0.369047634049
log_force_effectiveness_3	-0.427418292757
log_force_effectiveness_4	-0.370177496373

Physical representative

mass [kg]: 1.8729066173892384
CoG body [m]: [-0.00930364 0.0061865 -0.03619583]
force effectiveness: [0.67940464 0.69139248 0.65219069 0.69061174]
inertia [kg m²]:
[0.1432579143, 0.000849491363, -0.03607504209]
[0.000849491363, 0.1372737908, -0.006428741546]
[-0.03607504209, -0.006428741546, 0.2577515473]
principal moments [kg m²]: [0.13262393 0.13717583 0.2684835]

Per-bag raw residual-wrench RMS

failure_1_single: [0.37541305 0.16037403 0.69014599 0.06171192 0.09596269 0.10452244]